VMware to Proxmox VE Migration

Case Study

Case Study prepared by Illumia Solutions Pvt. Ltd.

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Executive Summary

Forceten Technologies Pvt. Ltd., a software developer providing service to Microfinance, NBFC, Co-operative societies, Rural bank and other Financial Institute, faced a critical inflection point in its IT infrastructure strategy. The rising cost of VMware licensing — a direct consequence of Broadcom's acquisition of VMware and the subsequent overhaul of its commercial model — made continuing with the existing virtualization platform financially untenable. With a firm organizational position of no new hardware procurement, and a live production environment hosting workloads with stringent downtime sensitivities, FTT needed a migration partner who could deliver platform transformation without operational disruption.

Forceten Technologies Pvt. Ltd. engaged **Illumia Solutions Pvt. Ltd. (ISPL)**, an official Proxmox partner with deep enterprise infrastructure expertise, to plan and execute a full-stack migration from VMware vSphere and Veeam Backup & Replication to **Proxmox Virtual Environment (Proxmox VE)** and **Proxmox Backup Server (PBS)**.

ISPL designed and delivered a phased, rolling migration strategy that systematically decommissioned each VMware host, commissioned it as a Proxmox VE node, and migrated workloads using Proxmox VE's built-in VMware migration tooling — all within the permissible downtime windows for each workload category. The existing SAN infrastructure was retained and integrated with the new platform, and the Veeam backup servers were repurposed as Proxmox Backup Server nodes, preserving the on-site and off-site backup architecture with encryption at rest applied to all backup data.

The outcomes of the engagement were unambiguous:

- **80% reduction in licensing costs**, delivering immediate and sustained financial relief.
- **Complete workload migration** with no VM lost in transition, including recovery of orphaned workloads from the Veeam backup repository.
- **Zero capital expenditure** — the entire transformation executed within the existing hardware footprint.
- **Strengthened data protection posture** through encrypted backup at rest across both on-site and off-site repositories.
- **Full compliance with all defined downtime thresholds**, eliminating operational risk throughout the migration lifecycle.

This engagement demonstrates that organizations in the financial sector need not choose between cost efficiency and operational resilience. With the right partner and a disciplined migration methodology, platform independence from high-cost proprietary vendors is both achievable and sustainable.

Background Information

Forceten Technologies Pvt. Ltd., a software developer providing service to Microfinance, NBFC, Co-operative societies, Rural bank and other Financial Institute, operated a business-critical IT infrastructure built around a multi-node **VMware vSphere** cluster. The virtualization environment hosted a significant number of Virtual Machines (VMs) supporting a diverse range of workloads — from mission-critical financial applications to supporting business functions — all of which demanded high availability and operational continuity.

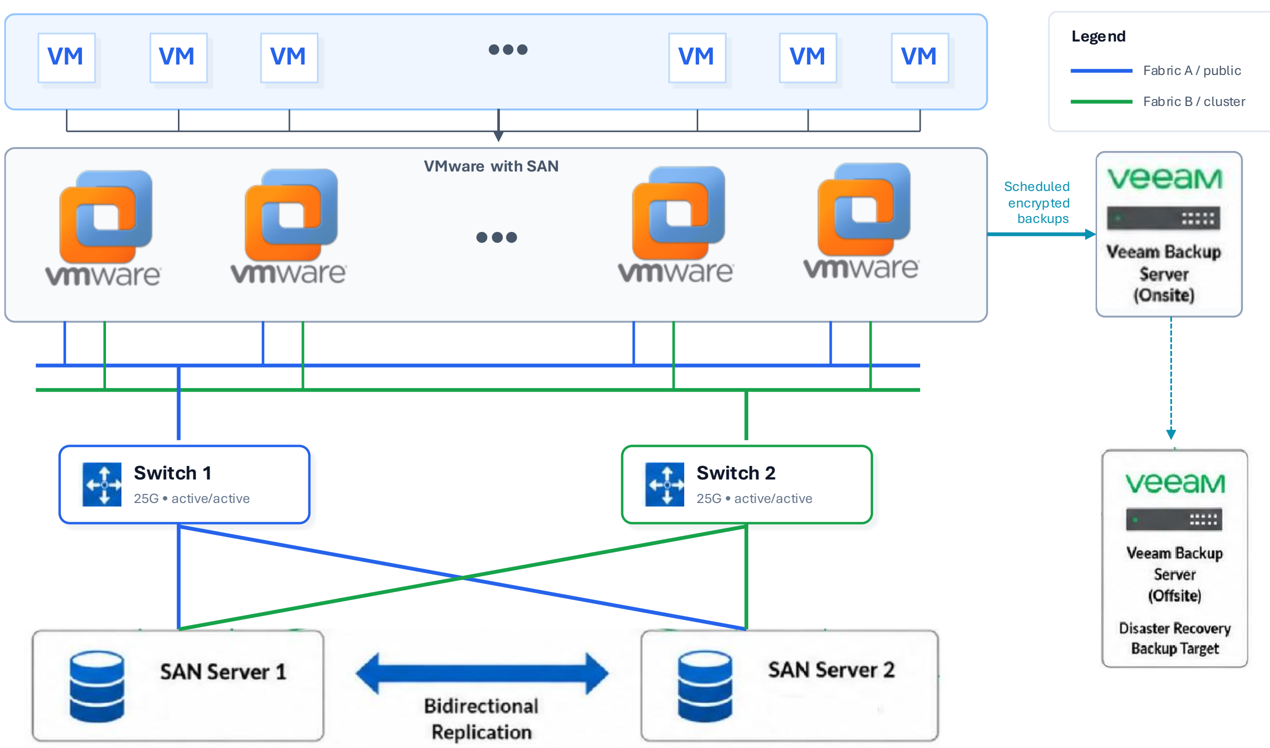
The storage backbone of the environment consisted of a **dual SAN (Storage Area Network)** setup — two SAN switches connected to two SAN servers — providing centralized, high-performance shared storage to the VMware cluster. This architecture is a well-established standard in enterprise financial environments where data integrity and storage reliability are non-negotiable.

For data protection, FTT had deployed **Veeam Backup & Replication** across two dedicated backup servers:

- An **on-site backup server** handling primary VM backups.
- An **off-site backup server** receiving replicated backups from the on-site server, serving as the disaster recovery copy.

The overall infrastructure represented a mature, enterprise-grade setup that had served FTT reliably. However, the environment was entirely dependent on VMware's licensing model — an arrangement that was about to become a significant financial concern.

Illumia Solutions Pvt. Ltd. (ISPL) was engaged as a trusted infrastructure partner, holding an active partnership with **Proxmox**, to lead the evaluation and execution of a platform transition.



Problem Statement

FTT's IT infrastructure, while technically sound, faced a confluence of challenges that collectively made the status quo financially and operationally unsustainable

1. Escalating VMware Licensing Costs
Following Broadcom's acquisition of VMware, the licensing model underwent a significant restructuring — moving away from perpetual licenses toward a subscription-based model with considerably higher price points. For FTT, this translated into a sharp and unavoidable increase in licensing expenditure, with no proportional gain in functionality or business value. Continuing with VMware was no longer a financially justifiable decision.

2. Backup Infrastructure Tied to the Existing Platform
FTT's data protection strategy was built entirely around Veeam Backup & Replication. While Veeam is a capable solution, it introduced an additional licensing overhead on top of the VMware costs. With the platform migration being inevitable, retaining Veeam as a standalone backup solution would mean carrying a redundant cost that no longer aligned with the new infrastructure direction.

3. Workload Continuity Risk During Migration
Not all workloads carried equal tolerance for downtime. The hosted workloads were broadly classified into three risk tiers:

Workload Category	Share of Total Workload	Permissible Downtime Window
Critical	40%	Maximum 1 hour (weekends, off-hours only)
Medium	30%	Up to 4 hours (weekends, off-hours only)
Low	30%	Up to 24 hours (weekends, off-hours only)

Exceeding these downtime thresholds carried real operational risk — including potential disruption to financial operations, downstream process failures, and reputational exposure. This made the migration not just a technical exercise, but a carefully governed operational event.

4. Hardware Constraint
FTT had a firm position of not procuring any new hardware for this transition. The entire migration and new platform commissioning had to be executed within the boundaries of the existing physical infrastructure — leaving no margin for error in capacity planning and sequencing.

5. Orphaned VM Workloads
A subset of VMs existed only within the Veeam backup repository and had no active presence in the VMware environment. These workloads still held business relevance and needed to be recovered and re-platformed onto the new infrastructure — adding an additional layer of complexity to the migration scope.

Taken together, these factors defined a migration project that demanded precise planning, phased execution, and zero tolerance for scope deviation — particularly given the financial sector context where operational disruptions carry consequences well beyond IT.

Solution Proposed

ISPL proposed a structured, phased migration strategy designed to transition FTT's entire virtualization and data protection infrastructure from VMware vSphere and Veeam to **Proxmox Virtual Environment (Proxmox VE)** and **Proxmox Backup Server (PBS)** — entirely within the existing hardware footprint, and without breaching any of the defined downtime thresholds.

1. Platform of Choice — Proxmox VE
Proxmox VE is an enterprise-grade, open-source virtualization platform built on the KVM hypervisor and LXC container technology. As an official Proxmox partner, ISPL recommended Proxmox VE as the ideal replacement for VMware vSphere for the following reasons:

- **Significant cost reduction** through elimination of per-host commercial licensing fees.
- **Native VMware migration tooling** built directly into the platform, reducing migration complexity and third-party dependencies.
- **Cluster-native architecture** supporting high availability and live migration across nodes.
- **Full compatibility** with FTT's existing SAN infrastructure, supporting iSCSI and Fibre Channel storage integration.
- **Active enterprise support** available through Proxmox's official subscription model, ensuring FTT would not be moving to an unsupported platform.

2. Phased, Rolling Node Migration
Given the no-new-hardware constraint and the tiered downtime sensitivity of the workloads, ISPL proposed a **rolling migration approach** — migrating the infrastructure node by node rather than as a bulk cutover. This approach ensured that:

- The VMware cluster remained operational throughout the migration, protecting critical and medium-priority workloads.
- Each Proxmox VE node was commissioned, validated, and workload-verified before the next VMware host was decommissioned.
- Workload migrations were scheduled strictly within the permissible downtime windows for each category — critical workloads being migrated last and with the most controlled conditions.

3. SAN Infrastructure Reuse
Rather than introducing new storage infrastructure, ISPL proposed leveraging the existing dual SAN setup. The available storage headroom — approximately 40% of total capacity — was sufficient to carve out dedicated LUNs for the Proxmox VE cluster without disrupting the active VMware workloads during the transition period. This directly honored FTT's hardware constraint while maintaining storage-level resilience throughout.

4. Recovery of Orphaned VMs
For the VMs present only in the Veeam backup repository, ISPL proposed restoring them directly from Veeam backups following Veeam's standard VM restore procedure and onboarding them into the Proxmox VE cluster as part of the migration scope. This ensured no workload was left behind during the platform

transition.

5. Backup Infrastructure Replacement — Proxmox Backup Server

ISPL proposed decommissioning both Veeam servers upon completion of the VM migration and repurposing the underlying hardware as **Proxmox Backup Server** nodes. The proposed backup architecture mirrored the existing design:

- A **primary on-site PBS node** with a dedicated datastore integrated with the Proxmox VE cluster for regular, scheduled VM backups.
- A **secondary off-site PBS node** configured to replicate backups from the primary datastore, maintaining the disaster recovery posture.
- **Encryption at rest** applied to all backup data — both on the primary and replicated copies — ensuring data security compliance consistent with the expectations of a financial sector environment.

Implementation

ISPL executed the migration in a series of carefully sequenced phases, each validated before proceeding to the next. Every activity was planned around FTT's workload downtime tiers and scheduled exclusively within the permitted off-hour weekend windows.

Phase 1 — Initial Proxmox VE Cluster Commissioning

The migration began without touching any active workload. ISPL first identified the VMware hosts running the lowest priority VMs and consolidated those workloads onto other hosts within the existing VMware cluster. This freed up two physical hosts without any impact on the broader environment.

These two hosts were then cleanly removed from the VMware vSphere cluster. Proxmox VE was installed and commissioned on both, and they were configured as the founding nodes of a new **2-node Proxmox VE cluster**.

On the storage side, the existing SAN infrastructure was assessed and new LUNs were carved out from the available storage headroom within the existing SAN servers. These LUNs were presented to and integrated with the Proxmox VE cluster, establishing a shared storage foundation consistent with FTT's existing enterprise storage architecture.

Phase 2 — Initial Workload Migration to Proxmox VE

With the 2-node Proxmox VE cluster operational and storage-backed, ISPL migrated the VMs that had previously resided on the two freed VMware hosts. Proxmox VE's **built-in VMware migration tool** was used for this purpose, eliminating the need for any third-party migration utilities.

An **offline migration method** was chosen — VMs were cleanly shut down prior to migration — to ensure data consistency and avoid any partial-state transfer risks. Once migrated, each VM was brought online individually and subjected to performance validation in coordination with the respective application owners. Only after sign-off was the next VM started.

Phase 3 — Rolling Node-by-Node Migration

Following the successful completion of Phase 2, ISPL repeated the same structured process for every remaining VMware host in the cluster:

1. Workloads on the target VMware host were migrated to other active hosts within the VMware cluster.
2. The host was removed from the VMware vSphere cluster.
3. Proxmox VE was installed and the host was joined to the growing Proxmox VE cluster.
4. VMs previously hosted on that node were migrated using the built-in VMware migration tool with offline migration.
5. Migrated VMs were validated with application owners before being declared production-ready.

This cycle continued until every VMware host had been decommissioned, migrated, and joined to the Proxmox VE cluster — at which point the VMware vSphere environment ceased to exist entirely.

All workload migrations were scheduled in strict alignment with the downtime tier classifications — critical workloads were migrated within their 1-hour window, medium workloads within 4 hours, and low-priority workloads within the 24-hour allowance — all during off-hour weekend slots.

Phase 4 — Recovery of Orphaned VM Workloads

Following the completion of all VMware-hosted VM migrations, ISPL addressed the subset of VMs that existed exclusively within the Veeam backup repository. These VMs were restored directly from the Veeam backup server following Veeam's standard VM restore procedure and brought online within the Proxmox VE cluster. Post-restoration validation confirmed that the recovered workloads were functioning as expected.

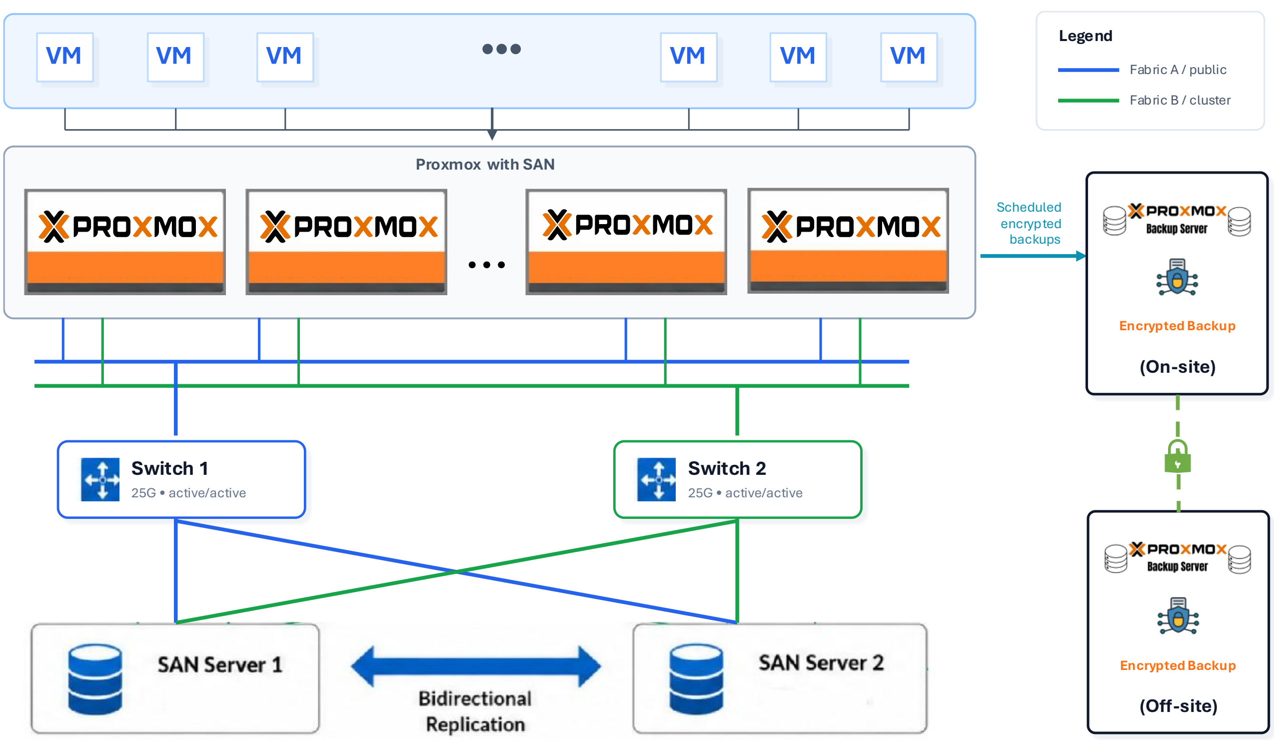
Phase 5 — Backup Infrastructure Decommissioning and Proxmox Backup Server Commissioning

With all VMs successfully running on the Proxmox VE cluster, ISPL proceeded to retire the Veeam infrastructure. Both the on-site and off-site Veeam backup servers were decommissioned and their underlying hardware was repurposed as **Proxmox Backup Server** nodes.

The implementation of the new backup architecture was as follows:

- A **primary PBS datastore** was created on the on-site backup server and integrated with the Proxmox VE cluster. Scheduled backup jobs were configured for all VMs, with **encryption at rest** applied to all backup data.
- The **off-site PBS node** was configured to replicate the primary datastore, maintaining the disaster recovery backup copy. Replicated data retained the same encryption at rest posture.

End-to-end validation of the backup and restore workflow was performed before the implementation was declared complete.



Conclusion

The successful completion of this migration project delivered a modernized, cost-efficient, and operationally resilient virtualization infrastructure for Forceten Technologies Pvt. Ltd. — without the procurement of a single piece of new hardware and without breaching any of the defined workload downtime thresholds.

What Was Achieved

By transitioning from VMware vSphere to Proxmox VE and replacing Veeam Backup & Replication with Proxmox Backup Server, FTT realized the following outcomes:

- **80% reduction in licensing costs** — the most immediate and quantifiable business outcome, directly addressing the financial pressure that initiated the project.
- **Full workload continuity** — every VM previously running in the VMware environment was successfully migrated to the Proxmox VE cluster, with no workload lost in transition.
- **Recovery of orphaned workloads** — VMs that existed only within the Veeam backup repository were successfully restored and onboarded into the new platform, closing a gap that predated the migration project itself.
- **Modernized data protection** — the Proxmox Backup Server deployment replicated FTT's existing backup architecture — on-site primary with off-site replication — while adding encryption at rest across all backup data, strengthening FTT's data security posture.
- **Zero new capital expenditure** — the entire transformation was executed within the existing hardware footprint, converting what could have been a capital-intensive project into a purely operational and service-driven engagement.

Broader Significance

For organizations in the financial segment, infrastructure decisions carry weight beyond IT operations. Licensing dependencies on a single vendor — particularly one that has undergone significant commercial restructuring — represent a strategic risk. This engagement demonstrated that a well-planned, phased migration to an open-source enterprise platform like Proxmox VE can deliver enterprise-grade reliability and security without the commercial constraints of proprietary virtualization stacks.

ISPL's role as an official Proxmox partner, combined with its deep infrastructure expertise, enabled FTT to navigate a complex, multi-phase migration with precision — protecting operational continuity at every step while achieving a transformative reduction in infrastructure overhead.

This project stands as a reference for financial sector organizations evaluating a similar transition — proving that platform independence, cost efficiency, and operational resilience are not competing objectives, but achievable together.

Hardware Infrastructure